Why PFAS half-lives differ between humans, rats and mice

A review of the evidence, the assumptions behind the published numbers, and what is still unresolved

Evidence base	186 full texts read; 24 per-paper extractions; 1,438 database rows; 61 references
Underlying review	report/REPORT.md, ~2,922 lines, 59 runnable scripts, 21 figures
Repository	github.com/sdey17/pfas_tk

Contents

1	Introduction: one identity, three questions
2	Evidence base and method
3	Where the species difference lives: distribution or clearance?
4	The mechanism: fractional renal reabsorption
5	Transporter identity, and why the mechanism does not transfer
6	The second loop: enterohepatic recirculation
7	Plasma protein binding, and a conflict that was not one
8	Does exposure level change elimination?
9	The human volume of distribution
10	Toward structure-activity: the renal handling ratio
11	What this review corrects in the published record
12	Coverage: how much of the field is empty
13	Open questions, and what would close them
14	Conclusions
	References

Abstract

Published serum half-lives for the same per- and polyfluoroalkyl substance differ by orders of magnitude between humans and laboratory animals, between rat and mouse, between male and female, and sometimes between papers describing the same experiment. This review asks where that variance actually sits. Because half-life is not a property but a ratio - the volume a chemical apparently occupies divided by the rate the body clears it - the question is answerable, and the answer is unambiguous: across six datasets that measure both terms in both sexes within single experiments, the volume of distribution varies 1.6-fold while clearance varies 56-fold and changes sign between species. The variance is in clearance.

One mechanistic axis then orders every species for which data exist. Fractional renal reabsorption runs from 99.94% in humans through 97% in the male mouse and 93.7% in the male rat to net tubular secretion in the female rat and the rabbit. Rodent half-lives follow from it within 1.7-fold over a 37-fold span, though that agreement establishes that renal clearance is nearly all of total clearance in rodents rather than validating an independent model. Two refinements matter. The axis is two factors rather than one: of the 333-fold male-mouse-to-human renal clearance gap, the escape fraction carries 50-fold and the six-fold lower human glomerular filtration rate carries 6.5-fold. And humans run a second, enterohepatic loop of comparable completeness, which closes the residual the renal axis leaves and explains why bile-acid sequestrants shorten human half-lives several-fold.

The review also audits the provenance of the numbers regulation uses. The human volume of distribution behind nine adopted clearance factors is a calculation rather than a measurement, reproducible to three figures from its source paper's own supplementary table given an assumed elimination rate drawn from a study in the same two communities; and a widely-inherited plasma free fraction is roughly 164-fold too high because a calculated lower bound was read as a measurement. Eighteen such corrections are documented, one of them to this work's own method.

Finally, half-life is argued to be the wrong endpoint for any structure-activity model, because it carries a body-size term. A dimensionless renal handling ratio is proposed in its place and evaluated on the only dataset that supports the comparison - a cocktail study whose absolute clearances and volumes both run two- to threefold below the one-compound-at-a-time literature while their ratio does not, a shared offset that leaves every between-compound contrast the endpoint is built on intact.

1. Introduction: one identity, three questions

Per- and polyfluoroalkyl substances persist in the body for a long time, and how long is the single number that drives their regulation: it converts an external dose into an internal one, and it is what any cross-species extrapolation has to carry. The published values do not agree. Human PFOA half-life estimates span roughly seventeen-fold, from about half a year to eight and a half; the male rat and the female rat differ by about seventy-fold in the same experiment [1, 2]; and the rat and the mouse, two rodents of similar size, differ in a direction that reverses between sexes [3].

Disagreements of that size usually mean a quantity is being compared across things that are not comparable. The way into the problem is that half-life is not an independent property. It is fixed by an identity:

$$t_{1/2} = ln2 \cdot V_d / CL$$

where V_d is the apparent volume of distribution - the number relating total body burden to plasma concentration, not an anatomical volume - and CL is clearance, the volume of plasma irreversibly cleared per unit time. Any two of the three terms fix the third. A study that measures serum decay obtains the rate constant directly and never needs a volume; a study that works from mass balance needs a volume and a complete excretion accounting, each uncertain severalfold, and both multiply into the answer. The practical consequence runs through this entire review: most published disagreements are not disagreements about measurements. They are differences in **which two terms were measured and which one was assumed** - and in the most consequential case, the assumed one is the number everybody quotes.

Three questions follow, and this review answers them in order. First, is the species difference a difference in how much is held or in how much is let go - distribution or clearance? Second, what did each reported half-life assume? Third, does exposure level itself change elimination, as saturable kinetics would predict?

2. Evidence base and method

The review rests on 186 full texts read directly, 24 per-paper extractions made from those texts, and a consolidated database of 1,438 rows across four schemas - toxicokinetic parameters, protein binding, transporter kinetics, and adopted regulatory values. Animal serum time courses come from the US EPA's compiled per-record database [4], which digitised or transcribed them from the primary studies - principally the monkey intravenous series [5, 6], the rodent dose series [7, 8, 9] and a low-dose chronic study [10]. Human values come from the primary papers and from agency assessments [11, 12, 13, 14, 15, 16].

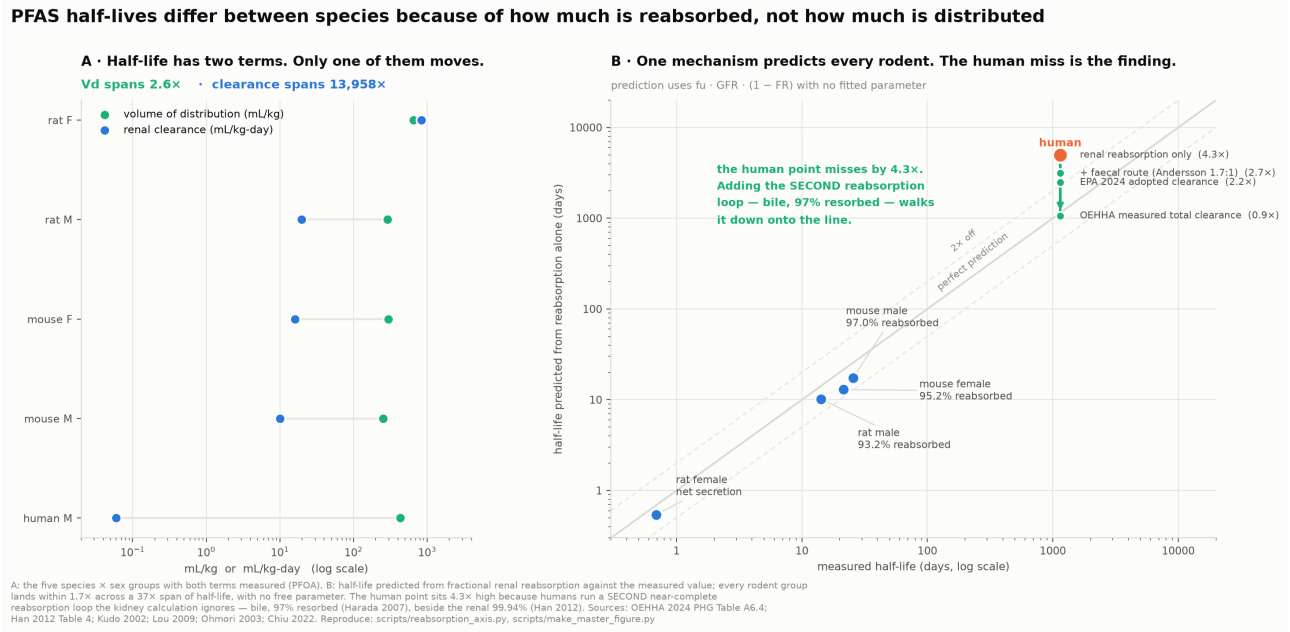
Three rules govern what is in the database, and they are the reason several of this review's conclusions differ from the secondary literature.

Read the table, not the sentence. Where a paper's abstract and its own tables disagree, the table wins and the discrepancy is recorded. Two of the largest corrections in section 11 are of exactly this kind, and so is the correction this review makes to its own earlier work.

Mark what was computed. Every row records whether a value was measured, fitted, assumed, or computed in this work, and carries a PMID or DOI and the table it came from. A clearance derived from an assumed volume is not the same kind of object as a clearance measured from urine, and conflating the two is how the problem in section 9 propagated.

Flag your own biases rather than deleting them. This review refitted the EPA raw curves and found its own fitting procedure biased on biphasic data. The affected column is kept, flagged, and documented, because it remains valid for the within-curve comparisons it is used for - and because deleting it would hide a failure mode that applies to much of the published literature too.

Figure 1



The argument in one pair of panels. Left: *across five species-by-sex groups in which both terms were measured, the volume of distribution spans 2.6× while renal clearance spans 13,958× - the variance is in clearance, not distribution (section 3).* ***Right:*** *half-life predicted from fractional renal reabsorption alone, against observation, on log axes with the 1:1 line. Every rodent point falls within 1.7× over a 37× span - which, as section 4 notes, shows that renal clearance is nearly all of total clearance in rodents rather than validating an independent model, since the reabsorbed fraction is derived from the same renal clearance. The human point sits 4.3× high, and the walk-down beneath it - renal only 4.3×, plus faecal 2.7×, EPA clearance 2.2×, OEHHA measured clearance 0.9× - is what led to the second reabsorption loop (section 6).* ***Note*** *that the panel annotations carry the widely-circulated OEHHA adaptation of the reabsorption table (male rat 93.2%, human 99.8%), while the table in section 4 carries the source values (93.7% and 99.94%). The discrepancy is not an error here - it is one of the corrections listed in section 11, left visible rather than silently harmonised. Regenerate with scripts/make_master_figure.py.*

3. Where the species difference lives: distribution or clearance?

The identity in section 1 makes this testable rather than rhetorical, provided both terms are measured in the same animals. Six datasets satisfy that condition - three species, two compounds, four laboratories, no cross-study comparison required [1, 2, 17, 18, 7, 8].

	V_d male/female	clearance female/male
span across all six datasets	**1.33 - 2.18×**	**0.78 - 44.3×**
	a 1.6× spread	a 56× spread

The distribution ratio is male-higher in every dataset and never leaves a narrow band, which is what one would expect of a quantity set mostly by body composition and plasma protein concentration. The clearance ratio ranges over nearly two orders of magnitude **and changes sign between species**. That asymmetry is the central finding of this review, and everything after it is an attempt to explain the clearance term.

A single recent experiment reproduces the result internally. Argoul and colleagues dosed eleven PFAS as one cocktail in female mice and fitted all of them simultaneously [19]; across those eleven compounds, clearance spans 5,254-fold while steady-state volume of distribution spans 7.7-fold. Same animals, same assay, same model. Note that this varies *chemistry* at fixed species and sex, where the table above varies sex at fixed chemistry, so it is corroboration by analogy rather than replication - though it is hard to explain why clearance should be the volatile term along both axes unless it simply is the volatile term. This study's *absolute* clearances run about 2.5× below the one-compound-at-a-time literature, for reasons set out in section 10; the comparison here is a ratio within one experiment, from which a shared scaling cancels, so the 5,254-fold against 7.7-fold is unaffected.

Taken from primary sources rather than compilations, the species gap is also concentrated in one sex. Female rat divided by female mouse PFOA clearance is 373-496×, against 7.0× in males [1, 20] - this comparison *is* cross-study, rat and mouse from different laboratories, and carries the usual caveat. Tatum-Gibbs and colleagues replicate the whole structure in a second compound with strains matched: a rat sex ratio of 21.9× against a mouse ratio near unity [3]. Any account of 'the species difference' that does not mention sex is describing an average over two quite different animals.

4. The mechanism: fractional renal reabsorption

PFAS are filtered freely at the glomerulus to the extent they are unbound, and then substantially recovered from the tubular filtrate. Renal clearance is therefore what escapes:

$$CL_{renal} = f_u \cdot GFR \cdot (1 - FR)$$

where f_u is the unbound fraction in plasma and FR the fraction reabsorbed. Han and colleagues assembled this across species [21]; the table below is read from that paper's Table 4 at source rather than from any adaptation of it.

species	sex	GFR (L/d/kg)	renal CL (mL/d/kg)	reabsorbed
human	both	2.57	0.03	**99.94%**
mouse	female	16.7	16	**95.2%**
mouse	male	16.7	10	**97.0%**
rat	female	14.4	666	**net secretion**
rat	male	14.4	18.2	**93.7%**
Japanese macaque	female	8.5	32	**81.2%**
Japanese macaque	male	8.5	15	**91.2%**
dog	female	5.3	50.8	**52%**
dog	male	5.3	43	**59%**
rabbit	female	4	670	**net secretion**
rabbit	male	4	640	**net secretion**

Read at source into db/primary_2026/han2012_table4_reabsorption_axis.csv. Two values differ from a widely-cited adaptation of this table; see section 11.

The ordering is the half-life ordering. A caveat on how much that demonstrates, because it is easy to overclaim: the reabsorbed fraction is *derived* from the measured renal clearance by rearranging the equation above, so predicting a half-life from it restates the identity $t_{1/2} = \ln 2 \cdot V_d / CL_{renal}$ rather than testing a reabsorption model independently. What the agreement to 1.7× over a 37× span does establish is that **renal clearance accounts for nearly all of total clearance in these rodents** - a real and non-trivial result, and the reason the human residual in section 6 stands out so sharply. The axis orders the species and names the quantity that varies; it does not independently predict them.

The reabsorbed fraction does get one independent check. A separate dataset, different laboratory and different animals, recomputes mouse PFOA reabsorption at 95.9% [19], inside the 95.2-97.0% range already in use.

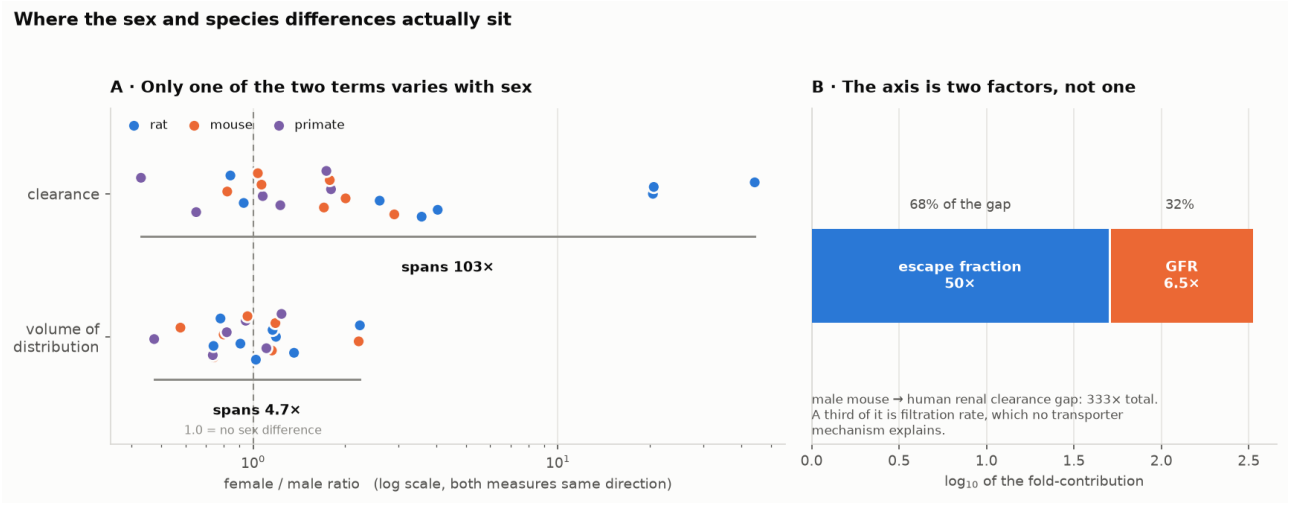
Elimination hands off from kidney to gut as the chain lengthens

The axis is renal, and for the shorter carboxylates that is nearly the whole story: PFHpA leaves the male rat 92% in urine over five days. By PFNA that falls to 2.0% and by PFDA to 0.2%, with faeces becoming the major route [22]. So a renal clearance is close to a total clearance at C7 and is a small minority of it at C10, and any human accounting that assigns one route fraction across compounds is wrong in a predictable direction. The handoff is also sex-dependent: female rats excrete 51% of a PFNA dose in urine where males excrete 2.0% [22], which is the same sex difference as section 3 seen through the route split rather than the rate.

The axis has two factors, not one

It is tempting to read the table as 'humans reabsorb more', and this review initially did. The absolute numbers say otherwise: humans recover 51 mL/d/kg of filtrate, against 270 in the male rat and 318-324 in the mouse. Humans reabsorb *less* in absolute terms and still clear far more slowly, because they filter far less to begin with. Decomposing the 333× male-mouse-to-human renal clearance gap gives 50× from the escape fraction and 6.5× from the six-fold lower human GFR - 68% and 32% on a log scale. Reabsorption is the larger term, which is why the single-axis framing works at all; but a third of the difference is filtration rate, and no transporter story explains that third.

Figure 2



A: *21 matched female/male pairs - eight compounds across three species - with both ratios written in the same direction, so the contrast cannot be an artefact of how each was expressed. The volume of distribution stays inside a 4.7× band straddling 1.0; clearance spans 103× and crosses it in both directions. This is a wider set than the six datasets in the table above, and it says the same thing more strongly.* ***B:*** *the male-mouse-to-human renal clearance gap split into its two multiplicative factors; because they multiply, their logarithms add, and the escape fraction takes 68% of the distance. Regenerate with scripts/make_review_figures.py.*

5. Transporter identity, and why the mechanism does not transfer

If reabsorption is the controlling step, some apical transporter must perform it, and in the rat the sex difference constrains which one. A candidate has to satisfy two conditions simultaneously: it must be sex-divergent in the right direction, and it must actually carry PFOA. Those two conditions are jointly much more restrictive than either alone. The table covers the transporters for which both properties have been measured, so it is an elimination among tested candidates rather than a proof that no untested protein qualifies.

transporter	sex-divergent?	transports PFOA?	verdict
Oatp1a1	**yes**, 23× male-predominant, androgen-induced	**yes**, C8-C10	**the only candidate satisfying both**
Oat2	yes, strongly	**no** - three negative reports, two species	excluded by transport
Oat1 / Oat3	no	yes	excluded by regulation

The transport evidence is from direct uptake assays [23, 24, 25] and the regulation evidence from expression studies under hormonal manipulation [1, 26, 27, 28]; the organic anion transporting polypeptides as a family have been shown to contribute to PFAA disposition in both humans and rats [29]. The 23× figure is from the primary source; a frequently quoted '2.5-fold' belongs to a different transporter (see section 11).

The surprise is in the human

The reabsorbed *fraction* transfers across species - that is what section 4 shows. The *mechanism* does not. OATP1A2, the closest human orthologue of rat Oatp1a1, does not mediate saturable PFOA uptake at all [30]. Human apical reabsorption instead runs through OAT4 and URAT1 [30, 31], neither of which is androgen-regulated - which is consistent with the absence in humans of anything like the rat's seventyfold sex difference. Extrapolating a rat mechanism to humans and extrapolating a rat reabsorbed fraction to humans are therefore very different acts, and only the second is supported.

6. The second loop: enterohepatic recirculation

The renal axis leaves a residual. Predicted human half-life comes out about 4.3× short of observation - visible as the single off-line point in Figure 1. Three candidate explanations exist: the human reabsorbed fraction is underestimated, the human clearance figure is wrong, or a second elimination route is being reabsorbed too.

The third is correct, and the number was already in an agency document. Harada and colleagues sampled serum and bile from four gallstone-surgery patients and measured a biliary resorption rate of 0.97 [32]. Bile PFOS (27.9 ng/mL) actually *exceeds* serum (23.2), so bile is a genuine excretion route - but 97% of what is secreted is recovered from the gut.

loop	fraction reabsorbed	source
renal (PFOA)	**99.94%**	Han 2012 Table 4 [21]
biliary (PFOS)	**97%**	Harada 2007, n = 4 [32]

Humans therefore run two near-complete reabsorption loops rather than one. Both are near unity, both lengthen the half-life, and - the testable part - **both are interruptible**. If a 97% resorption loop is blocked, elimination should accelerate several fold, and it does. In a treatment programme using cholestyramine and plasma donation, apparent PFOS half-life was 1.2 y in nineteen treated participants against 7.3 y in nine observed ones, and PFHxS 2.5 y against 9.4 y [33]. Supporting evidence comes from a cholestyramine series [34] and a cross-over trial reporting 63% lowering in twelve weeks against 3% in controls [35]. Plasma donation alone, which removes burden without touching either loop, produces a much smaller effect [36].

The loop also settles an apparent contradiction in the human excretion literature. One study measuring faeces under ongoing intake finds faecal elimination dominant for PFOS [37]; another, following a single labelled dose, barely detects it [38]. At 97% resorption, **gross** biliary flux is large while **net** faecal elimination is small, so the two studies measured different quantities and both are right. The caveat is that the constant doing this work rests on four patients.

7. Plasma protein binding, and a conflict that was not one

The unbound fraction f_u enters the reabsorption equation directly, so its value propagates into everything above. The literature appeared to contain a hundredfold disagreement: a long-standing result reports PFOA 'over 90% bound' to

plasma protein [39], implying f_u near 0.1, while recent measurements at environmentally realistic ligand:protein ratios give 0.00061 [40].

Reading both papers at source dissolves the conflict. The '>90%' is not a measurement but a *calculation* from a dissociation constant and an albumin concentration, and as a lower bound it is satisfied by 0.00061 as comfortably as by 0.1. The underlying difference in dissociation constant is a titration artefact: the older work titrated 50-60 µM albumin with 0.1-3 mM PFOA, a ligand:protein ratio of 1.7:1 to 60:1, against 0.004:1 or below in the newer work - while human serum sits at 10^{-5} to 10^{-3}:1. Near-saturating ratios understate affinity, and structural work on the albumin complex is consistent with multiple sites of differing affinity [41].

The defect was in the inheritance rather than in either measurement. Physiologically based models read '>90% bound' as 'about 90% bound' and adopted a free fraction roughly **164× too high**. Section 10 shows how far that single substitution propagates.

Turning a binding constant into something a mass balance can use

A mass-balance model does not want an affinity, it wants a capacity: how many litres of water's worth of PFAS one kilogram of albumin holds. For albumin at concentration P with n equivalent sites of dissociation constant K_d, Langmuir occupancy gives bound ligand $n \cdot P \cdot C_w/(K_d + C_w)$ per litre of solution, and dividing by the sorbent mass $P \cdot M_{alb}/1000$ makes P cancel. In the trace limit that leaves $\log D = \log K_a + \log n - 1.823$ on a mass basis, or −1.688 on the volume basis Qin uses. Association constant and partition coefficient are therefore strictly proportional, with a log-log slope of exactly one. What the mass balance needs is the intercept, and the intercept is set by the molar-mass basis and by n - neither of which an affinity measurement usually reports. Moving n from one site to six shifts log D by 0.78 without changing any measurement.

The conversion is not a guess. Qin's own albumin term, $C_{max}/(K_d + C_w)$ plus a non-specific constant, is this expression, which identifies C_{max} as n times the molar concentration of albumin inside the pure albumin phase. Qin reports C_{max} both per litre and per mole of albumin, and the ratio recovers that concentration to within 0.3% for all four measured compounds. It also reveals that their fitted n is below one site per albumin for PFBA and PFOS: lumped effective parameters, not site counts.

Run backwards the conversion becomes a diagnostic. Asking what stoichiometry reconciles the tightest reported binding constant with [40]'s directly measured partition coefficient gives n = 1.01 for PFOA and 1.12 for PFHpA: the two kinds of measurement agree at one binding site with nothing fitted. For the short chains the implied n runs to 20-48, meaning the available constants are far too weak to reproduce the measured partitioning, and for PFOS and PFBS it falls to 0.36 and 0.02, meaning they are too tight. Those are disagreements about the chemistry rather than about stoichiometry, and no conversion will reconcile them.

This is why proportionality is not enough to tune a model. Writing *a* for albumin's share of total sorptive capacity, an error of factor *f* in the albumin partition coefficient moves every predicted concentration by $1/[(1-a) + f \cdot a]$. The error is bounded by $1/(1-a)$ - a model cannot put more than the whole dose in the cell - but it scales with *a*, which is not a constant:

at 100 µM	a	if D_alb is 10× low	ceiling
PFBA	0.19	1.21×	1.2×
PFOA	0.89	5.06×	9.2×
PFNA	0.94	6.51×	16.8×

Built by scripts/kd_to_partition.py.

Across the three assays at a fixed compound *a* moves by under 3%, because all three run 10% serum with the same cell line. The axis it runs along is chain length. So one multiplicative error in the albumin parameter produces a 1.2× error for PFBA and a 6.5× error for PFNA in the same experiment, and a single fitted correction factor cannot absorb a discrepancy that varies that way. Serum content is where the assay axis bites instead: at 0% serum the albumin parameter is irrelevant, at 10% it carries nearly nine tenths of the capacity, so a model validated serum-free says nothing about its behaviour at 10%. The practical reading is that for long chains in these assays the ceiling is 7-17×, so albumin is where the leverage is, and it has to be fixed as an absolute value per compound on the basis the model actually uses.

The substitution was tried, and it failed informatively

With the conversion in hand the obvious experiment is to give both mass-balance models the measured albumin value instead of their own and re-run the 29 cell-uptake points. Since the assays use fetal bovine serum, [40]'s bovine column applies. It sits below both models for every compound - a seventh of Qin's value for PFHxA, a quarter for PFOA - so substituting it shrinks the albumin sink and leaves more of the dose for the cell, and both models already over-predict what the cell takes up.

Predicted over measured, geometric mean across the 29 points:

variant	bias	|err|	at ≤25 µM	at 100 µM	dose swing
Qin, published	3.43×	3.95×	1.58×	5.52×	3.49×
Qin + measured albumin	4.74×	4.77×	4.78×	4.72×	**0.99×**
Armitage, published	**2.33×**	**2.42×**	2.73×	2.12×	0.78×
Armitage + measured albumin	5.09×	5.10×	5.79×	4.71×	0.81×

Built by scripts/mbm_fischer_albumin.py.

Both models get worse. One thing improves, and it is structural: Qin's error swings 3.5× across the dose range as published and is flat once the albumin coefficient stops saturating. The dose dependence Qin produces is its own Langmuir albumin term coming off saturation, not a statement about how PFAS enter cells.

Inverting the balance asks whether albumin could carry the error at all. The albumin coefficient that would make each model unbiased is in every case *above* what the model already uses, while the measurement is *below* it - the data and the residual point in opposite directions. For PFHxA the value needed exceeds anything ever measured for that compound by eight to thirteenfold, even allowing six binding sites per albumin. So albumin is where the sensitivity is and not where the error is, which leaves the cell term, the plastic term and the measured concentrations themselves. That conclusion is not reachable by comparing proportionality; it needs the absolute conversion.

Which leaves the cell

The two models disagree about what a cell is more sharply than about anything else. Armitage's is 79% water for the shortest chain and essentially pure membrane lipid for the longest, with protein contributing a fifth of one percent throughout, because only 3.5% of protein is treated as sorptive and that fraction is modelled as octanol. Qin's is 92-95% protein at every chain length and carries no water phase at all. They differ on the cell partition itself by up to ninefold, yet their predicted fraction of dose in the cell agrees within about twofold, because the albumin term dominates the denominator and damps the difference. Agreement on the output is not corroboration of the mechanism.

Chain-length dependence of the cell partition, decades per CF_2, from four independent routes:

source	slope
uptake data, by inverting the balance	+0.55
Qin model	+0.52
Armitage model	+0.78
Ryu 2024 whole-tissue unbound fraction	0.46-0.53
Qin Table 2, cell homogenate	**+0.18**

Built by scripts/mbm_cell_term.py.

Measured against Qin's own cell-homogenate numbers both models look three to four times too steep, which reads as a model defect until the other two routes are added: the cellular uptake data and whole-tissue dialysis both land on the models' slope, and Qin's Table 2 sits alone, twentyfold above what the uptake requires for PFBA. Substituting those measured values confirms it from the other side - the mean bias improves, Armitage's to 1.25×, while the absolute error gets worse and the shortest chain is over-predicted twentyfold. Centring a fit while destroying its trend is what substituting an outlier looks like.

So the chain-length dependence is not the defect, and what remains is a level error in the cell term of several fold, worst in the middle of the series. That is the right size and the right direction to be the whole over-prediction, where albumin was neither. Two candidates remain: the sorptive capacity assigned to cell protein, where Qin's measured non-specific affinity and Armitage's octanol surrogate are wildly different guesses at the same quantity and are the substance of each model's cell term, and the cell volume the whole result is normalised by, which is uncertain to threefold across the three cell

compositions in use.

Species and structure, from the studies that hold method fixed

Most of the 741-row binding file cannot answer a comparative question, for the reason above: a single affinity number per pair does not exist. What can are the few studies that fix the method and vary one thing.

On **species**, [42] measured four albumins by one assay. Dissociation constant in mM, lower is tighter:

compound	F-carbons	human	rat	bovine	porcine	rat / human
PFBA	3	2.64	1.90	3.06	1.28	0.72×
PFBS	4	1.68	1.41	0.80	1.25	0.84×
GenX	4	1.57	1.64	2.55	2.12	1.04×
PFHxA	5	1.71	1.33	2.17	0.95	0.78×
6:2 FTSA	6	0.41	0.33	1.39	0.52	0.80×
PFOA	7	0.79	0.40	0.99	0.37	**0.51×**
PFOS	8	0.69	0.42	0.86	0.38	0.61×

Built by scripts/binding_species_and_structure.py from db/protein_binding.csv.

The study reports the interspecies differences as statistically significant for every bound compound, and they are - but the magnitudes are small. Across all four species the highest-to-lowest ratio never exceeds about fourfold and is usually under twofold. Rat and porcine albumin bind the carboxylates and sulfonates more tightly than human, the ether acid less tightly, in the study's own words and in the table. Albumin is a conserved protein and behaves like one. The direction is again unhelpful for explaining kinetics: rat albumin holds PFOA twice as tightly as human while the rat clears it about a thousandfold faster. A second method agrees - [40] measured human and bovine albumin for 27 compounds at a physiological ligand:protein ratio and the bovine-minus-human difference in log D is under 0.3 log units for most of them, median -0.14. Bovine albumin is a fair stand-in for human, which matters because much of the literature uses it.

On **structure**, [40]'s ten chain lengths give three results, and only the first is the textbook one. **Chain length is not monotonic.** The carboxylates rise from 2.16 at four fluorinated carbons to a maximum of 5.18 around ten to eleven, then fall to 3.87 at thirteen - a slope of +0.45 log units per CF_2 on the way up and -0.44 on the way down. A structure-activity model linear in carbon number is wrong at both ends. An independent dataset reaches the same shape: the QSAR of [43] over the [42] affinities finds perfluoroalkyl acids of seven to nine carbons bind human albumin most tightly and those above ten bind less, and its docking gives a mechanism - the tightest binders form a hydrogen bond with Arg348 and salt bridges to Arg348 and Arg485, while the weakest form neither. **The head group is worth about a log unit** at matched chain length, with similar slopes rather than a different trend: PFHxS 4.77 against PFHpA 4.10 at six carbons, PFHpS 5.49 against PFOA 4.48 at seven. **And albumin stops being the main carrier.** γ-globulin binds two to three log units weaker at short chain lengths, closes the gap steadily, and at thirteen carbons binds more tightly than albumin does. Treating protein binding as albumin binding degrades exactly where it matters, since the long chains are the persistent ones. [44] reviews the same picture mechanistically: the fluorinated chain occupies the binding cavity while the acid head group hydrogen-bonds to residues, so chlorine substitution, ether insertion and branching all act through chain length and hydrophobicity rather than independently.

Three findings were added to the file after the primary papers were obtained, and each changes a conclusion rather than adding a row. **A dissociation constant quoted without its binding site is as incomplete as one quoted without its stoichiometry.** [45] measured five perfluoroalkyl acids at three sites on human albumin with three probes, and regulatory summaries carry the tryptophan column, which the authors identify as the weakest of the three. For PFOS, Sudlow site II binds 345× more tightly than the quoted tryptophan value - a dissociation constant of 0.13 µM rather than 45 µM. The same paper supplies the control the field usually lacks: octanoic acid binds site II at 1.1×10^7 M^{-1}, tighter than every PFAS measured there, so these compounds occupy the fatty-acid sites without outcompeting the fatty acids.

Ionic strength moves affinity in one direction, and buffers are usually too dilute. [46] is the only chain-length series measured across added salt: ten PFAS on bovine albumin with sodium chloride stepped from 0 to 500 mM. Affinity falls monotonically, about 1.5 to 2× between a 20 mM buffer and the roughly 150 mM of plasma, and the shift is independent of chain length and head group. Constants measured in dilute buffer therefore overstate binding before any other correction.

And whole-tissue binding is no more a species difference than albumin binding is. [47] dialysed sixteen PFAS against brain, heart, kidney, liver and lung homogenate from human, Wistar Han rat, CD-1 mouse and C57BL/6J mouse - 320 unbound fractions, all of them now in the database. Taking the ratio of rodent to human within each compound and tissue, the three rodents land at geometric means of 0.93, 0.92 and 0.80 of the human value, with the worst single cell between a quarter and three times human. Rat liver predicts human tissues at an R^2 of 0.98 or better. Liver binding sits within twofold of the other tissues for fourteen to sixteen of the sixteen compounds in every organism, despite the liver's enrichment in liver fatty-acid binding protein, which argues against that protein as a driver of retention - the opposite reading to [48], whose thermal-shift assay finds a 7.7× human-over-mouse spread for that protein in isolation. The two measure different things, affinity for one purified protein against the net sorptive capacity of whole homogenate, but they cannot both carry the species difference, and the whole-tissue design is the one closer to the in vivo question.

That dataset also carries the only side-by-side measurement of the two mouse strains, which matters because the mouse limb of section 3 rests on a C57BL/6 study while the older literature is CD-1. Ratio of CD-1 to C57BL/6J unbound fraction, geometric mean over the sixteen compounds:

tissue	CD-1 / C57BL/6J	range	over 2-fold
liver	1.20	0.63-1.76	0 / 15
brain	1.05	0.53-1.98	0 / 16
heart	1.01	0.49-2.61	3 / 16
kidney	**1.58**	0.61-3.28	2 / 16
lung	1.01	0.52-1.64	0 / 16

Built by scripts/ryu_funbound_species_tissue.py from db/ryu2024_funbound.csv.

Kidney is the only tissue with both a shifted centre and a failing equivalence test in the paper's own statistics: its two one-sided tests find human against C57BL/6, CD-1 against C57BL/6 and rat against C57BL/6 kidney not equivalent within twofold, while every other tissue passes for every pair. Two cautions against over-reading it. This is tissue binding, and renal clearance is driven by plasma unbound fraction times glomerular filtration, which this study did not measure, so the finding bears on renal retention rather than on filtration. And a 1.6× strain difference is small beside the roughly thousandfold rat-to-human clearance gap. It is a real, kidney-specific strain effect in the right organ and nowhere near large enough to be the confounder on its own.

The same batch of papers also doubled the differential scanning fluorimetry coverage: [49] reports affinities for 65 PFAS across 20 structural classes plus three fatty acids, of which 19 did not bind albumin at all - every fluorotelomer alcohol, acrylate and methacrylate among them. Where it overlaps [42] the two agree to a geometric mean of 0.98× across seven compounds, so the low affinities that assay reports are reproducible. They are a property of reading a constant off a melting transition, not a disagreement about chemistry, which is why they rank congeners well and should not be carried into a kinetic model as affinities.

One caution about that file now that it holds 741 rows: a single study supplies 320 of them. Any aggregate taken across the whole table is an aggregate of [47] with a margin of everything else, so group by study before summarising.

The gap is in the field, not in this file. There is still no in vitro measurement of PFAS binding to *mouse* serum albumin. A PubMed search returns one record, and it measures mouse albumin concentration rather than binding. So the albumin species question is answerable for human, rat, bovine and porcine and not for the mouse - the animal whose clearance and unbound fraction section 10's endpoint is built from, and whose f_u therefore has no in vitro albumin measurement to decompose it against.

What binding does to elimination, quantified

Only unbound PFAS is filtered, so the clearance free filtration alone would produce is $f_u \cdot GFR$, and the measured renal gap between species factorises into binding, body size and transport. For PFOA the female-mouse-to-human renal clearance gap is 120× whatever is assumed; what the assumption changes is how that gap is apportioned.

human f_u	binding	body size	transport	dominant term
0.00061 measured [40]	**14.3×**	3.9×	2.2×	**binding**
0.02 assumed [21]	0.4×	3.9×	70.7×	transport
0.10 inherited	0.1×	3.9×	353.5×	transport

Mouse limb from [19] (CL_{renal} 3.59 mL/d/kg, f_u 0.87%), human from [21] Table 4 (0.03 mL/d/kg). Built by scripts/binding_elimination_decomposition.py.

Under the measured value **binding is the largest single term**: humans hold PFOA about fourteenfold more tightly than mice, which by itself keeps fourteenfold less of it in front of the glomerulus. Under the assumed values the apportionment inverts - binding runs the wrong way, implying humans bind less tightly than mice, and transport has to carry 71 to 354-fold to balance. The mouse and human free fractions come from different methods at different ligand:protein ratios, which is the artefact this section diagnoses, so the fourteenfold is the least secure number in the table; and the mouse limb carries the cocktail study's low clearance bias, which lands on the transport term and would raise 2.2× to about 5×, still far below 71×.

Binding alone does not explain long human half-lives, and the numbers say why. At f_u = 0.00061 free filtration would clear 1.57 mL/d/kg, which with V_d = 170 mL/kg is a half-life of **75 days** - weeks, not years. Measured human renal clearance is 0.03 mL/d/kg, **52× lower still**, because tubular reabsorption recovers **98.1%** of what is filtered. Binding sets how much reaches the glomerulus; reabsorption disposes of almost all of it. Binding is the species-variable factor and reabsorption the near-complete one in every species measured, so a model with binding right and reabsorption wrong still misses the half-life by two orders of magnitude.

8. Does exposure level change elimination?

If reabsorptive transport saturates, elimination should accelerate at high burden, half-life should fall with dose, and a single clearance factor could not be transferred between a contaminated community and the general population. The evidence is weak but not null, and it splits informatively by the level at which the comparison is made.

comparison	direction
within a person over time	supports dose dependence
within a species across doses (rat refits)	supports, weakly: slope about +0.11
between people in a cohort	**contradicts** - the lowest-exposure tertile declines fastest
between water districts	slope above the mechanistic ceiling of 1, i.e. bias

Within-unit comparisons support concentration dependence and between-unit ones do not, which is the signature of a between-unit confounder - and the source study names it: age [50]. Re-running that paper's own variance decomposition, age carries 2-13× the partial R^2 of initial PFAS concentration, and the exposure-tertile effect is 25-37% the size of the age effect. Older people eliminate more slowly and have higher accumulated burdens, so the between-person association is a positive saturation effect minus a larger negative age effect.

Across volume of distribution the slopes run -0.21 to +0.25 with inconsistent sign, i.e. no relationship. The one clean exception runs the other way: the female rat shows a slope of +0.21 over a 3,200× dose range against -0.02 in the male, which is what saturable *secretion* looks like - consistent with the female rat sitting at the secretory end of the axis in section 4.

Whether human exposures reach saturation is unresolved and is taken up in section 13.

9. The human volume of distribution

Nine of the 42 adopted regulatory clearance factors catalogued here are computed from a single human volume of distribution, 170 mL/kg [51]. That number is not a measurement.

Its source paper's supplementary table is headed *'Input data for the calibration of the Vd parameter'* with a final column headed *'calculated Vd'*, and its published values reproduce to three figures from intake, serum concentration and an assumed elimination rate. The assumed rate was taken from a study that measured half-life in **the same two communities** [52], which is what makes the construction circular rather than merely indirect.

The consequence is sharper than circularity, and it is asymmetric. Forming a clearance from the identity makes the half-life cancel exactly:

half-life assumed	V_d that follows	derived clearance
2.3 y (the value used)	**168 mL/kg**	0.132-0.138 mL/kg-day
3.8 y	**277 mL/kg**	0.132-0.138 mL/kg-day (*unchanged*)

Clearance reduces to dose divided by serum concentration and is robust; the volume does not cancel and is proportional to whatever half-life was assumed. So adopting '170' alongside a different half-life silently contradicts the data it came from. The arithmetic is reproduced from the paper's own supplementary table to within 1.000-1.003× in scripts/thompson_vd_circularity.py. The paper's own corrigendum is an instance of exactly this error, corrected by a factor of 0.6 - the ratio of two rate constants.

Independent measurements do not support the adopted value in either direction: direct estimates cluster at 74, 121 and 113-199 mL/kg [37, 38, 36], all *below* the assigned range, while the one population pharmacokinetic fit gives 430 mL/kg [53], far above it. A mass-balance half-life anchored on 170 mL/kg [54] moves to within 5% of the population fit when the fitted volume is substituted instead. That is one study, so it does not show the whole seventeen-fold span to be a single assumed constant; what it shows is that the mass-balance end of the span - the end that needs a volume at all - collapses toward the serum-decay end when the volume is changed.

10. Toward structure-activity: the renal handling ratio

A long-term aim of this work is to relate PFAS toxicokinetics to molecular structure. Half-life cannot be that endpoint. Because it is $\ln 2 \cdot V_d/CL$, it carries a glomerular filtration term that differs 6.5× between mouse and human with no change in chemistry whatever; a structural model fitted to it is being asked to absorb body size, and it cannot.

The replacement proposed here is dimensionless - measured renal clearance divided by the clearance free filtration alone would produce:

$$R = CL_{renal} / (f_u \cdot GFR)$$

R below 1 is net reabsorption, R above 1 net secretion, R = 1 pure filtration. It divides out GFR, so species of different size become comparable, and divides out f_u, so the binding step is not counted twice; what remains is the transport step, which is the part a structural model could plausibly learn. It is the axis of section 4 re-expressed, since R = 1 - FR, on a scale that neither crowds against a ceiling at 1 nor runs unboundedly negative under secretion.

Only one dataset supports the comparison: nine compounds with both renal clearance and unbound fraction measured in one experiment, one species, one sex and one laboratory - female mice [19]. Note that this is a different animal from the male mouse of section 4's table, whose reabsorption figures come from a different source [21]. R spans 875× across them.

compound	F-carbons	head group	ether O	f_u %	$\log_{10}$ R	handling
PFDA	9	-COOH	0	0.42	-2.509	reabsorbed
PFHxS	6	$-SO_3H$	0	1.3	-2.344	reabsorbed
PFNA	8	-COOH	0	0.35	-1.766	reabsorbed
PFOA	7	-COOH	0	0.87	-1.384	reabsorbed
PFOS	8	$-SO_3H$	0	0.25	-1.305	reabsorbed
PFBA	3	-COOH	0	77.0	-0.993	reabsorbed
GenX	5	-COOH, ether	1	26.0	-0.985	reabsorbed
PFO2OA	5	-COOH, ether	2	10.0	0.186	**SECRETED**
PFHxA	5	-COOH	0	25.0	0.433	**SECRETED**

Built by scripts/qsar_endpoint_table.py into db/qsar/qsar_endpoint_table.csv. 'F-carbons' counts carbons bearing fluorine, which excludes a carboxylate's acid carbon and includes every carbon of a sulfonate - a definition chosen so that PFNA and PFOS come out the same size, which the data can then falsify.

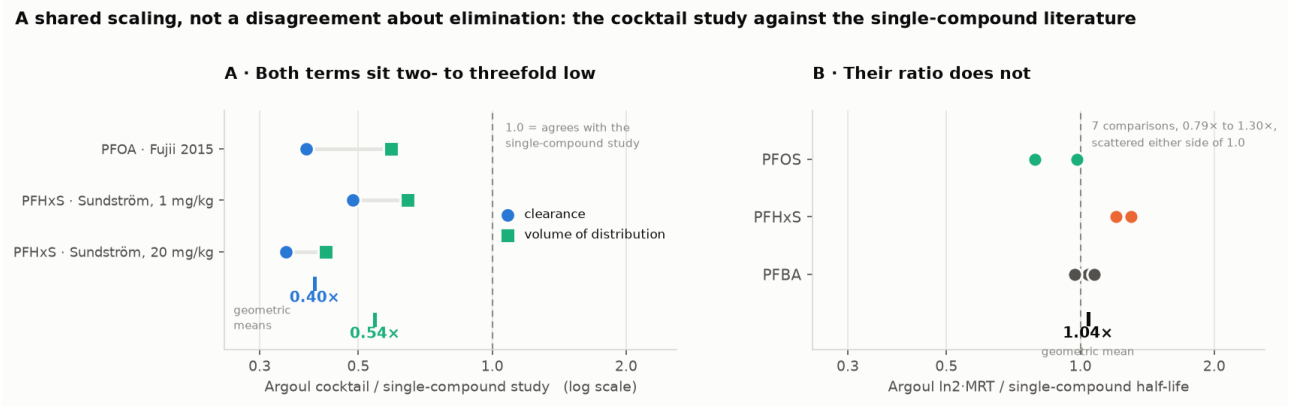
Three results constrain any structure-activity model built on this endpoint. **Chain length alone does not order it**: within the carboxylates the series is non-monotonic and PFHxA crosses into net secretion, so a model using carbon number as its sole descriptor is already falsified on nine compounds. **The head group carries about threefold at matched chain**

length - PFNA against PFOS, both with eight fluorinated carbons, differ 2.9×, so the equivalence the descriptor assumed is a good approximation but not free. **Ether oxygens move the endpoint 1.4 log units at constant chain length, and non-monotonically**: PFHxA with none is secreted, GenX with one is reabsorbed, PFO2OA with two is secreted. The replacement chemicals sit on both sides of the divide, which is a regulatory observation as much as a chemical one.

How far the one dataset can be trusted

Everything above rests on a single experiment, and that experiment is unusual in three ways at once: eleven PFAS dosed as one cocktail, at 0.019-1.55 mg/kg, fitted by nonlinear mixed effects. Each of those could bias the absolute numbers, so the comparison against the one-compound-at-a-time literature is worth making explicitly. All comparators below are female mice, matching [19].

Figure 4



A: *every single-compound female-mouse study that reports both clearance and volume, each on its own row, against the cocktail. The grey connector joins the two parameters within a study. Geometric means 0.40× for clearance [55, 17] and 0.54× for volume.* ***B:*** *the same comparison for half-life, which is their ratio. Seven comparisons across three compounds [56, 17, 6], spanning 0.79× to 1.30×, geometric mean 1.04×. Only comparisons with one attributable dose are shown: three studies pooled two dose levels into a single reported parameter [20, 3, 13] and are excluded, which costs PFOA, PFNA and PFHxA. Including them gives ten half-life comparisons at 0.99× rather than seven at 1.04×, so nothing turns on the exclusion. Built by scripts/argoul_vs_single_compound.py into db/argoul_vs_single_compound.csv; drawn by scripts/make_review_figures.py.*

One caveat on how much of a check the clearance and volume rows are. In most of these studies the volume is not measured apart from the clearance: non-compartmental analysis gives $V_z = CL/\lambda_z$ and $V_{ss} = CL \cdot MRT$. For the two Sundström rows 'clearance low and volume low' is largely one observation seen twice. Fujii is the exception - volume from dose/C(0) by back-extrapolation, clearance from dose/AUC - and there the two are low independently, 0.38× and 0.59×. One row is thin evidence, so the claim defended here is the narrower one: the rate constant agrees across the half-life comparisons and the scale does not.

The shape of that result is more informative than either half of it. Clearance and volume are both low by about twofold, and the half-life is not low at all - which is what a shared multiplicative factor on both terms looks like, because in $\ln 2 \cdot V_d/CL$ such a factor cancels. Four candidates fit, and testing the most obvious one weakens it. **Dose** looks the natural candidate - section 8's saturable-reabsorption account predicts clearance rising with dose, and the cocktail was dosed below its comparators - but Argoul tuned the dose per compound (0.019-1.55 mg/kg, tracking persistence rather than sweeping a range), so the gap must be matched compound by compound, and then it fails to order the discrepancies. Fujii dosed PFOA at 0.128 mg/kg against the cocktail's 0.080, a gap of 1.6×, and disagrees *most* (0.38×); Lou at 1 mg/kg and above, a gap of at least 12×, agrees *best* (0.75×). Dropping the cocktail from the PFHxS dose series and extrapolating Sundström's own two doses down to it predicts 2.04 mL/kg/d against 1.30 observed, so dose closes part of that gap and about 1.6× survives it. Dose is a contributor, not the explanation. **Strain** is a confound the comparison cannot remove: Argoul dosed C57BL/6 mice while every comparator is CD-1 except Fujii, which is FVB/NJcl, so all thirteen comparisons cross strains and no study in hand dosed the same compounds in both under matched conditions. It is listed without a direction because none can be inferred here; the comparison file carries both strains per row. **Cocktail competition** for reabsorptive transporters would raise clearance, not lower it, so it argues against the observed direction - unless the competition is for plasma binding sites, which would raise f_u and lower the apparent volume. **Mixed-effects shrinkage** compresses spread rather than shifting the centre, so it is the weakest of the four.

What this does and does not cost the endpoint above. R is built from an absolute renal clearance, so a 2.5-fold low bias in that clearance is a 2.5-fold low bias in R - a flat -0.40 log unit offset on every compound. Because it is flat, it changes

nothing that this section actually claims: the ranking, the 2.9-log-unit span, the non-monotonicity in chain length and the head-group and ether contrasts are all differences between compounds, and a shared factor cancels from every one of them. It does move the compounds relative to the fixed divide at R = 1, but not far enough to matter - the nearest reabsorbed compound sits 0.98 log units below the divide, so the bias would have to be 9.6× rather than 2.5× to reclassify anything. What it does cost is the cross-species comparison in the next subsection, where an absolute mouse value is set against an absolute human one; read those gaps as carrying a factor of about 2.5 on top of the f_u uncertainty, in the direction of understating the mouse.

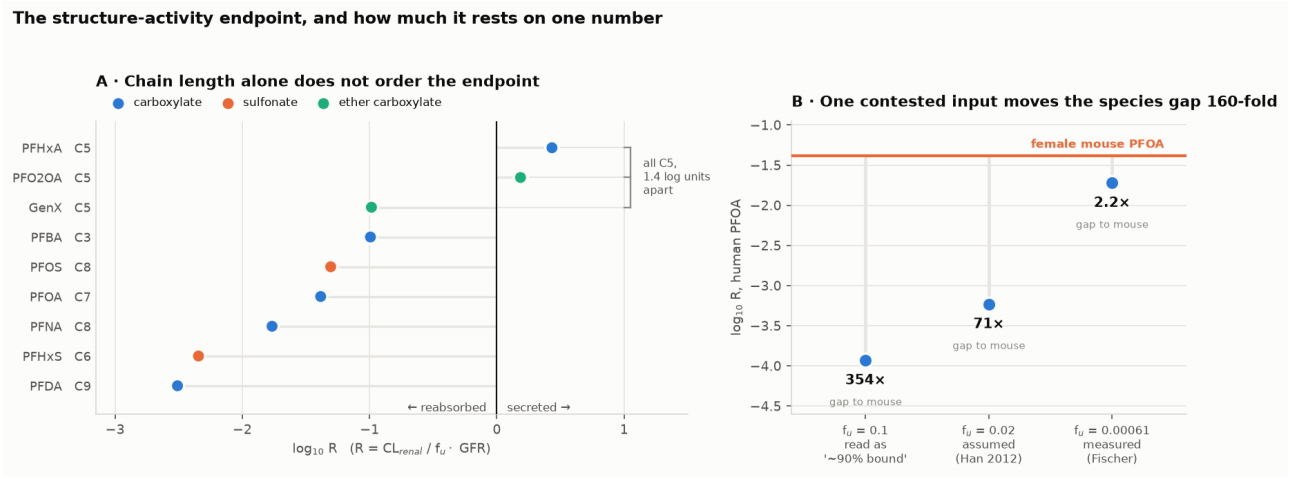
A sensitivity that bears on the species question

R is linear in $1/f_u$, so it inherits the uncertainty of section 7 in full. For human PFOA, with renal clearance and GFR both fixed, the three free fractions in circulation give three different answers:

f_u	source	$\log_{10}$ R	gap to female mouse
0.10	models reading '>90% bound' as 'about 90%'	-3.93	354×
0.02	the stated assumption in [21]	-3.23	71×
0.00061	measured at physiological ligand:protein [40]	-1.72	**2.2×**

Under the assumed value the female-mouse-to-human difference in the transport step is about seventyfold and the species gap is transport biology. Under the measured value it is about twofold, and the species gap is almost entirely **binding** - which would mean the transporter literature of section 5 is explaining a quantity that barely differs between the species. This is stated as a sensitivity rather than a result, because the mouse and human free fractions come from different methods at different ligand:protein ratios, which is precisely the artefact section 7 diagnoses. It is also why the experiment proposed in section 13 is the one worth doing first. To size the problem plainly: the endpoint spans 2.9 log units across the nine compounds a structure-activity model would be fitted to, and the disagreement over this one input moves a single compound by 2.2 of them. The endpoint is well posed; the inputs are not yet good enough to fit it.

Figure 3



***A:** the nine compounds on the proposed endpoint, ordered by it, coloured by head group and labelled with fluorinated-carbon count. The vertical rule at zero is the reabsorption/secretion divide. The bracket marks the three compounds with five fluorinated carbons, which span 1.4 log units - the clearest single refutation of chain length as a sole descriptor. **B:** human PFOA on the same endpoint under each of the three unbound fractions in circulation, against the female mouse. The quantity a structure-activity model would have to explain spans 2.9 log units across panel A; the disagreement about one input moves one compound by 2.2 of them. Regenerate with scripts/make_review_figures.py.*

11. What this review corrects in the published record

Each entry is traced to the table or figure that contradicts it; the full list of eighteen is section 8 of report/REPORT.md.

source	correction
US EPA [11]	the rat Oatp1a1 male/female ratio quoted as '2.5-fold' is a *different transporter's* number (OAT-K); the primary value is **23×** [1]
OEHHA [14]	its adaptation of the reabsorption table altered two values (human 99.94 to 99.8, male rat 93.7 to 93.2) relative to the source [21]

source	correction
Provenance	the f_u = 0.02 behind the axis is the source paper's assumption, not the agency's, and that paper calls its own values 'rough estimates' [21]
Andersson 2025 [37]	transposes the PFOA and PFOS volumes of [51] (it is PFOA 170, PFOS 230); an earlier claim in this work that [54] did so was itself wrong and is withdrawn
Cheng 2006 [57]	its abstract calls renal Oatp1a1 'female-predominant', then reports androgens *increase* it and concludes androgens are the exclusive cause - self-contradictory, and against [26, 27]
Metabolic inertness	'PFAS are metabolically inert' fails for 6:2 Cl-PFESA, which is biotransformed [58, 59]
This review	**its own refits of the EPA raw curves compress sex ratios about 13×**, as below

The self-correction matters most

This work refitted terminal slopes directly from the EPA raw curves [4], selecting the window with the best adjusted R^2. On a biphasic curve that criterion selects the shallow terminal tail. Tested against a published value for the same experiment, the fit returns **1.44 d against 0.08 d** for the female rat and compresses a **71× sex ratio to 5.6×** [1]. The bias runs one way, so every conclusion above survives and several strengthen - but that column must not be read as comparable to published half-lives. It carries a flag and is kept rather than deleted, because it remains valid for within-curve comparison and because the same failure mode applies wherever a terminal slope is fitted without inspecting the window.

12. Coverage: how much of the field is empty

A review that only reports what is known overstates the state of the field, so the gap was measured. Crossing 39 chemicals by 11 species by 4 parameters gives 1,716 cells.

	cells
no data at all	**1,466 (85%)**
resting on a single study	94
PFOS, the best-covered compound in the world	29 of 44 filled
by species: human / rat / mouse / monkey / dog	71 / 61 / 45 / 28 / **2**

Mining four further compilations [15, 16, 60, 61] moved coverage from 143 to 156 cells, so the hole is real rather than a search artefact. The dog column deserves particular notice: it is the species that sits furthest down the reabsorption axis among the mammals with data, and it rests on two cells, one of which traces to a book chapter that could not be obtained.

13. Open questions, and what would close them

13.1 The unbound fraction, measured one way across species

This is the cheapest decisive experiment the review identifies, and it is first because two separate questions turn on it. Every value of R in section 10 is proportional to f_u, and the mouse and human numbers currently come from different methods at different ligand:protein ratios. Measuring them by one method at physiological ratio decides whether the species difference is binding or transport - a 33× swing in the answer - and simultaneously supplies the denominator any structure-activity model needs. It requires plasma, a dialysis or ultrafiltration rig and a mass spectrometer. No animals.

13.2 A measured human transport-affinity constant

Whether reabsorptive transport saturates at real human exposures depends on which parameter family one believes. Six independent in vitro half-saturation constants for PFOA against human transporters span 47 to 310 µM, from two laboratories, agreeing within a factor of seven [30, 25, 31]. The transport-affinity constant fitted inside physiologically based models is 0.133 µM - **354 to 2,336× lower than anything ever measured in a cell**. The in vitro values put human serum far below half-saturation; the fitted value puts three of four human populations at or above it, which would make clearance dose-dependent in contaminated communities and mean a single clearance factor cannot transfer between exposure settings. The weight of evidence has moved onto the in vitro side - the fitted values are fitted to plasma curves rather than measured, the mouse row carries standard errors exceeding its estimates, and the dose-response evidence of section 8 agrees with the in vitro answer - but nobody has measured a human value, and one measurement would close it outright.

13.3 Whether the biliary resorption constant replicates

The 0.97 of section 6 is now load-bearing for the human limb, and it rests on four surgical patients in a single study reached through an agency document [32]. A replication in any species under a stated protocol would be worth more than its cost.

13.4 The mouse renal Oatp1a1 sex ratio

The rat value is 23× [1]; the mouse equivalent is not established, and the mouse sex difference runs the opposite way to the rat's [3]. One quantitative PCR experiment would discriminate between the live hypotheses - though if the answer to 13.1 is that the species gap is binding, this question loses most of its force, which is why it is fourth rather than first.

13.5 Unobtained sources

Three documents would materially change specific numbers and could not be obtained: the dog toxicokinetic chapter underlying the dog column, an unpublished contract report on protein binding, and the primary rat dose series [9], which supplies roughly 70% of the rat PFOA observations used here only through its digitisation by [4].

14. Conclusions

The species difference in PFAS elimination is a clearance difference, not a distribution difference, and it is concentrated in females. One mechanistic axis - the fraction of filtered compound recovered from the tubule - orders every species for which both terms have been measured, and accounts for rodent half-lives within 1.7×. That axis is two factors rather than one, roughly two thirds reabsorption and one third glomerular filtration rate, and in humans it is joined by a second, enterohepatic loop of comparable completeness whose interruption shortens human half-lives several-fold.

The reabsorbed fraction transfers between species; the protein performing the reabsorption does not. That distinction matters for how animal data should be read across to humans: the quantitative axis is extrapolable, the mechanism is not.

Against that, two of the most widely inherited human parameters do not mean what their users take them to mean. The volume of distribution behind nine adopted clearance factors is a calculation whose assumed half-life came from the same communities it was calibrated on, and a plasma free fraction in general use is roughly 164× too high because a calculated lower bound was read as a measurement. Neither is a disputed measurement; both are provenance failures, and both are fixable by reading the source tables.

Finally, half-life should not be the endpoint of a structure-activity model, because it carries body size. The dimensionless renal handling ratio proposed here removes both the filtration and the binding terms, spans 875× across the nine compounds that support the comparison, and shows immediately that chain length alone will not do: the head group carries threefold at matched length and ether substitution moves the endpoint 1.4 log units non-monotonically, placing the replacement chemicals on both sides of the reabsorption-secretion divide. Extending that comparison to a second species, with unbound fraction measured the same way in both, is the single step that would turn this argument into a model.

Data and code availability

All extractions, the consolidated 1,438-row database, the 59 analysis scripts that regenerate every figure and table above, and the full review are at github.com/sdey17/pfas_tk. Every database row carries a PMID or DOI, the table it was read from, and whether its value was measured, fitted, assumed or computed in this work. Animal serum time courses are redistributed from the EPA compilation [4] under its MIT licence.

References

Numbered by order of first citation. Entries marked as held through a secondary source were not obtained in the original; the review does not quote them as if they had been.

1. Kudo N, Katakura M, Sato Y, Kawashima Y (2002). Sex hormone-regulated renal transport of perfluorooctanoic acid. *Chemico-Biological Interactions* 139(3):301-316. doi:10.1016/s0009-2797(02)00006-6
2. Ohmori K, Kudo N, Katayama K, Kawashima Y (2003). Comparison of the toxicokinetics between perfluorocarboxylic acids with different carbon chain length. *Toxicology* 184(2-3):135-140. doi:10.1016/s0300-483x(02)00573-5
3. Tatum-Gibbs K, Wambaugh JF, Das KP, Zehr RD, Strynar MJ, Lindstrom AB, Delinsky A, Lau C (2011). Comparative pharmacokinetics of perfluorononanoic acid in rat and mouse. *Toxicology* 281(1-3):48-55. doi:10.1016/j.tox.2011.01.003
4. Zurlinden TJ, Dzierlenga MW, Kapraun DF, Ring C, Bernstein AS, Schlosser PM, Morozov V (2025). Estimation of species- and sex-specific PFAS pharmacokinetics in mice, rats, and non-human primates using a Bayesian hierarchical methodology. *Toxicology*

and Applied Pharmacology 499:117336. doi:10.1016/j.taap.2025.117336. Data: github.com/USEPA/CPHEA-Animal-PFAS-PK

5. Butenhoff JL, Kennedy GL, Hinderliter PM, Lieder PH, Jung R, Hansen KJ, Gorman GS, Noker PE, Thomford PJ (2004). Pharmacokinetics of perfluorooctanoate in cynomolgus monkeys. *Toxicological Sciences* 82(2):394-406. doi:10.1093/toxsci/kfh302

6. Chang S-C, Noker PE, Gorman GS, Gibson SJ, Hart JA, Ehresman DJ, Butenhoff JL (2012). Comparative pharmacokinetics of perfluorooctanesulfonate (PFOS) in rats, mice, and monkeys. *Reproductive Toxicology* 33(4):428-440. doi:10.1016/j.reprotox.2011.07.002

7. Dzierlenga AL, Robinson VG, Waidyanatha S, DeVito MJ, Eifrid MA, Gibbs ST, Granville CA, Blystone CR (2020). Toxicokinetics of PFHxA, PFOA and PFDA in male and female Hsd:Sprague Dawley SD rats following intravenous or gavage administration. *Xenobiotica* 50(6):722-732. doi:10.1080/00498254.2019.1683776

8. Huang MC, Dzierlenga AL, Robinson VG, Waidyanatha S, DeVito MJ, Eifrid MA, Granville CA, Gibbs ST, Blystone CR (2019). Toxicokinetics of perfluorobutane sulfonate, perfluorohexane-1-sulphonic acid, and perfluorooctane sulfonic acid in male and female Hsd:Sprague Dawley SD rats after intravenous and gavage administration. *Toxicology Reports* 6:645-655. doi:10.1016/j.toxrep.2019.06.016, with its corrigendum, *Toxicology Reports* 8:365. doi:10.1016/j.toxrep.2021.02.001

9. Kemper RA (2003). Perfluorooctanoic acid: toxicokinetics in the rat. Unpublished report, DuPont-7473; US EPA public docket AR-226-1499. Haskell Laboratory, E.I. du Pont de Nemours. (Not obtained; its data are used as digitised by the EPA database [4].)

10. Iwabuchi K, Senzaki N, Mazawa D, Sato I, Hara M, Ueda F, Liu W, Tsuda S (2017). Tissue toxicokinetics of perfluoro compounds with single and chronic low doses in male rats. *Journal of Toxicological Sciences* 42(3):301-317. doi:10.2131/jts.42.301

11. US EPA (2024). *Final Human Health Toxicity Assessment for Perfluorooctanoic Acid (PFOA)*, EPA-815R24006, and its Appendix (Table B-26).

12. US EPA (2024). *Final Human Health Toxicity Assessment for Perfluorooctane Sulfonic Acid (PFOS)*.

13. US EPA (2023). *IRIS Toxicological Review of Perfluorohexanoic Acid (PFHxA)*.

14. California OEHHA (2024). *Public Health Goals for PFOA and PFOS in Drinking Water*, Appendix Tables A6.3, A6.4 and 4.8.1.

15. ATSDR (2021). *Toxicological Profile for Perfluoroalkyls*, Tables 3-5 and 3-6.

16. EFSA CONTAM Panel (2020). Risk to human health related to the presence of perfluoroalkyl substances in food. *EFSA Journal* 18(9):6223. doi:10.2903/j.efsa.2020.6223

17. Sundström M, Chang S-C, Noker PE, Gorman GS, Hart JA, Ehresman DJ, Bergman Å, Butenhoff JL (2012). Comparative pharmacokinetics of perfluorohexanesulfonate (PFHxS) in rats, mice, and monkeys. *Reproductive Toxicology* 33(4):441-451. doi:10.1016/j.reprotox.2011.07.004

18. Kim S-J, Heo S-H, Lee D-S, Hwang IG, Lee Y-B, Cho H-Y (2016). Gender differences in pharmacokinetics and tissue distribution of 3 perfluoroalkyl and polyfluoroalkyl substances in rats. *Food and Chemical Toxicology* 97:243-255. doi:10.1016/j.fct.2016.09.017

19. Argoul CML, Toutain P-L, Picard-Hagen N, Mselli-Lakhal L, Dauwe Y, Roques BB, Lacroix MZ, Gayrard V (2026). Nonlinear mixed-effects modeling of the intravenous and oral kinetics of eleven perfluoroalkyl substances in female mice. *Environmental Research* 303:124802. doi:10.1016/j.envres.2026.124802

20. Lou I, Wambaugh JF, Lau C, Hanson RG, Lindstrom AB, Strynar MJ, Zehr RD, Setzer RW, Barton HA (2009). Modeling single and repeated dose pharmacokinetics of PFOA in mice. *Toxicological Sciences* 107(2):331-341. doi:10.1093/toxsci/kfn234

21. Han X, Nabb DL, Russell MH, Kennedy GL, Rickard RW (2012). Renal elimination of perfluorocarboxylates (PFCAs). *Chemical Research in Toxicology* 25(1):35-46. doi:10.1021/tx200363w

22. Kudo N, Suzuki E, Katakura M, Ohmori K, Noshiro R, Kawashima Y (2001). Comparison of the elimination between perfluorinated fatty acids with different carbon chain length in rats. *Chemico-Biological Interactions* 134(2):203-216. doi:10.1016/s0009-2797(01)00155-7

23. Yang C-H, Glover KP, Han X (2009). Organic anion transporting polypeptide (Oatp) 1a1-mediated perfluorooctanoate transport and evidence for a renal reabsorption mechanism of Oatp1a1 in renal elimination of perfluorocarboxylates in rats. *Toxicology Letters* 190(2):163-171. doi:10.1016/j.toxlet.2009.07.011

24. Weaver YM, Ehresman DJ, Butenhoff JL, Hagenbuch B (2010). Roles of rat renal organic anion transporters in transporting perfluorinated carboxylates with different chain lengths. *Toxicological Sciences* 113(2):305-314. doi:10.1093/toxsci/kfp275

25. Louisse J, Pedroni L, van den Heuvel JJMW, Rijkers D, Leenders L, Noorlander A, Punt A, Russel FGM, Koenderink JB, et al. (2024). In vitro and in silico characterization of the transport of selected perfluoroalkyl carboxylic acids and perfluoroalkyl sulfonic acids by human organic anion transporter 1 (OAT1), OAT2 and OAT3. *Toxicology* 509:153961. doi:10.1016/j.tox.2024.153961

26. Cheng X, Maher J, Chen C, Klaassen CD (2005). Tissue distribution and ontogeny of mouse organic anion transporting polypeptides (Oatps). *Drug Metabolism and Disposition* 33(7):1062-1073. doi:10.1124/dmd.105.003640

27. Cheng X, Klaassen CD (2009). Tissue distribution, ontogeny, and hormonal regulation of xenobiotic transporters in mouse kidneys. *Drug Metabolism and Disposition* 37(11):2178-2185. doi:10.1124/dmd.109.027177

28. Buist SCN, Klaassen CD (2004). Rat and mouse differences in gender-predominant expression of organic anion transporter (Oat1-3; Slc22a6-8) mRNA levels. *Drug Metabolism and Disposition* 32(6):620-625. doi:10.1124/dmd.32.6.620

29. Zhao W, Zitzow JD, Weaver Y, Ehresman DJ, Chang SC, Butenhoff JL, Hagenbuch B (2017). Organic anion transporting polypeptides contribute to the disposition of perfluoroalkyl acids in humans and rats. *Toxicological Sciences* 156(1):84-95. doi:10.1093/toxsci/kfw236

30. Yang C-H, Glover KP, Han X (2010). Characterization of cellular uptake of perfluorooctanoate via organic anion-transporting polypeptide 1A2, organic anion transporter 4, and urate transporter 1 for their potential roles in mediating human renal reabsorption of perfluorocarboxylates. *Toxicological Sciences* 117(2):294-302. doi:10.1093/toxsci/kfq219

31. Louisse J, et al. (2023). Perfluoroalkyl substances (PFASs) are substrates of the renal human organic anion transporter 4 (OAT4). *Archives of Toxicology*. doi:10.1007/s00204-022-03428-6

32. Harada K, et al. (2007). Biliary excretion and enterohepatic recirculation of perfluorooctane sulfonate in humans. Values used here are as tabulated by US EPA (2016), *Health Effects Support Document for Perfluorooctane Sulfonate*; the original was not obtained.

33. Delaere I, Harris K, Gaskin S, Tefera Y, Mitchell K, Springer D, Mills S (2025). Changes in serum perfluorooctane sulfonic acid and perfluorohexane sulfonic acid concentrations in firefighters accessing a voluntary PFAS reduction treatment program. *Environment International* 202:109609. doi:10.1016/j.envint.2025.109609

34. Genuis SJ, et al. (2010). Human elimination of perfluorinated compounds under cholestyramine administration. (Cited via [37] and the EPA assessment; the original was not obtained.)

35. Møller S, et al. (2024). Cholestyramine cross-over trial for PFAS lowering. *Environment International*. doi:10.1016/j.envint.2024.108471 (cited via [37])

36. Gasiorowski R, Forbes MK, Silver G, Krastev Y, Hamdorf B, Lewis B, et al. (2022). Effect of plasma and blood donations on levels of perfluoroalkyl and polyfluoroalkyl substances in firefighters in Australia: a randomized clinical trial. *JAMA Network Open* 5(4):e226257. doi:10.1001/jamanetworkopen.2022.6257

37. Andersson AG, et al. (2025). The relative importance of fecal and urinary excretion of perfluorooctane sulfonic acid and perfluorooctanoic acid after high exposure: an observational study in Ronneby, Sweden. *Environmental Research* 285:122487. doi:10.1016/j.envres.2025.122487

38. Abraham K, Mertens H, Richter L, Mielke H, et al. (2024). Kinetics of 15 PFAS after a single oral dose in one adult volunteer: terminal half-lives, clearances and derived volumes of distribution. *Environment International*. doi:10.1016/j.envint.2024.109047

39. Han X, Snow TA, Kemper RA, Jepson GW (2003). Binding of perfluorooctanoic acid to rat and human plasma proteins. *Chemical Research in Toxicology* 16(6):775-781. doi:10.1021/tx034005w

40. Fischer FC, et al. (2024). Protein binding of PFAS measured by solid-phase microextraction at environmentally relevant PFAS:protein ratios. *Environmental Science & Technology*. doi:10.1021/acs.est.3c07415, and Fischer FC, et al. (2025). doi:10.1021/acs.est.5c05473

41. Maso L, Trande M, Liberi S, Moro G, Daems E, Linciano S, et al. (2021). Unveiling the binding mode of perfluorooctanoic acid to human serum albumin. *Protein Science* 30(4):830-841. doi:10.1002/pro.4036

42. Starnes HM, Jackson TW, Rock KD, Belcher SM (2024). Quantitative cross-species comparison of serum albumin binding of per- and polyfluoroalkyl substances from five structural classes. *Toxicological Sciences* 199(1):132-149. doi:10.1093/toxsci/kfae028

43. Gallagher A, Kar S, Sepúlveda MS (2023). Computational modeling of human serum albumin binding of per- and polyfluoroalkyl substances employing QSAR, read-across, and docking. *Molecules* 28(14):5375. doi:10.3390/molecules28145375. (Computational; its affinity data are the [42] set.)

44. Zhao L, Teng M, Zhao X, Li Y, Sun J, Zhao W, Ruan Y, Leung KMY, Wu F (2023). Insight into the binding model of per- and polyfluoroalkyl substances to proteins and membranes. *Environment International* 175:107951. doi:10.1016/j.envint.2023.107951. (Review.)

45. Chen Y-M, Guo L-H (2009). Fluorescence study on site-specific binding of perfluoroalkyl acids to human serum albumin. *Archives of Toxicology* 83(3):255-261. doi:10.1007/s00204-008-0359-x

46. Gao Y, Kibbey TCG, Lee MJ, O'Carroll DM (2025). Predicting salt effects on the binding of PFAS to protein. *Environmental Science & Technology* 59(49):26716-26728. doi:10.1021/acs.est.5c08779

47. Ryu S, Burchett W, Zhang S, Jia X, Sadegh Modaresi SM, Agudelo Areiza J, Rodrigues D, Zhu H, Sunderland EM, Fischer FC, Slitt AL (2024). Unbound fractions of PFAS in human and rodent tissues: rat liver a suitable proxy for evaluating emerging PFAS? *Environmental Science & Technology* 58(33):14641-14650. doi:10.1021/acs.est.4c04050

48. Han J, Fu J, Sun J, Hall DR, Yang D, Blatz D, Houck K, Ng C, Doering J, LaLone C, Peng H (2021). Quantitative chemical proteomics reveals interspecies variations on binding schemes of L-FABP with perfluorooctanesulfonate. *Environmental Science & Technology* 55(13):9012-9023. doi:10.1021/acs.est.1c00509

49. Starnes HM, Green AJ, Reif DM, Belcher SM (2025). An in vitro and machine learning framework for quantifying serum albumin binding of per- and polyfluoroalkyl substances. *Toxicological Sciences* 203(1):67-78. doi:10.1093/toxsci/kfae124

50. Li Y, Andersson A, Xu Y, Pineda D, Nilsson CA, Lindh CH, Jakobsson K, Fletcher T (2022). Determinants of serum half-lives for perfluoroalkyl substances after end of exposure to contaminated drinking water, Ronneby cohort. (Held as a structured abstract; full text paywalled.)

51. Thompson J, Lorber M, Toms L-ML, Kato K, Calafat AM, Mueller JF (2010). Use of simple pharmacokinetic modeling to characterize exposure of Australians to perfluorooctanoic acid and perfluorooctane sulfonic acid. *Environment International* 36(4):390-397. doi:10.1016/j.envint.2010.02.008, with its corrigendum, *Environment International* 36(6):652. doi:10.1016/j.envint.2010.05.008

52. Bartell SM, Calafat AM, Lyu C, Kato K, Ryan PB, Steenland K (2010). Rate of decline in serum PFOA concentrations after granular activated carbon filtration at two public water systems in Ohio and West Virginia. *Environmental Health Perspectives* 118(2):222-228. doi:10.1289/ehp.0901252

53. Chiu WA, et al. (2022). Bayesian estimation of human population toxicokinetics of PFOA, PFOS, PFHxS and PFNA from studies of contaminated drinking water. *Environmental Health Perspectives* 130(12):127001. doi:10.1289/EHP10103

54. Zhang Y, Beesoon S, Zhu L, Martin JW (2013). Biomonitoring of perfluoroalkyl acids in human urine and estimates of biological half-life. *Environmental Science & Technology* 47(18):10619-10627. doi:10.1021/es401905e

55. Fujii Y, Niisoe T, Harada KH, Uemoto S, Ogura Y, Takenaka K, Koizumi A (2015). Toxicokinetics of perfluoroalkyl carboxylic acids with different carbon chain lengths in mice and humans. *Journal of Occupational Health* 57(1):1-12. doi:10.1539/joh.14-0136-OA. (Values reached via OEHHA 2024 PHG Table 4.8.1 and ATSDR 2021 Table 3-6.)

56. Chang S-C, Das K, Ehresman DJ, Ellefson ME, Gorman GS, Hart JA, Noker PE, Tan Y-M, Lieder PH, Lau C, Olsen GW, Butenhoff JL (2008). Comparative pharmacokinetics of perfluorobutyrate in rats, mice, monkeys, and humans and relevance to human exposure via drinking water. *Toxicological Sciences* 104(1):40-53. doi:10.1093/toxsci/kfn057

57. Cheng X, Maher J, Lu H, Klaassen CD (2006). Endocrine regulation of gender-divergent mouse organic anion-transporting polypeptide (Oatp) expression. *Molecular Pharmacology* 70(4):1291-1297. doi:10.1124/mol.106.025122 (abstract only; full text paywalled)

58. Yi S, et al. (2022). Biotransformation of 6:2 chlorinated polyfluoroalkyl ether sulfonate in the rat. (Held as a structured abstract; see db/primary_2026/yi2022_clpfesa_rat.csv.)

59. Shi Y, Vestergren R, Xu L, Zhou Z, Li C, Liang Y, Cai Y (2016). Human exposure and elimination kinetics of chlorinated polyfluoroalkyl ether sulfonic acids (Cl-PFESAs). *Environmental Science & Technology* 50(5):2396-2404. doi:10.1021/acs.est.5b05849

60. US EPA (2025). *IRIS Toxicological Review of Perfluorohexanesulfonic Acid (PFHxS)*, Table 3-3.

61. New Jersey Drinking Water Quality Institute (2017-2018). Health-based maximum contaminant level support documents for PFOA, PFOS and PFNA.